Linear Mixed Models

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Warmup

Work on the warmup activity (handout).

Warmup

Data on prices and other information for 1561 Airbnb rentals in Chicago. Our goal is to model the relationship between price and overall satisfaction score.

- `price`: The price in US dollars of a one night rental
- `overallSatisfaction`: an average satisfaction score, given as numbers between 2.5 and 5
- `neighborhood`: the neighborhood in which the rental is located. There are 43 neighborhoods, and 25 - 50 rentals per neighborhood

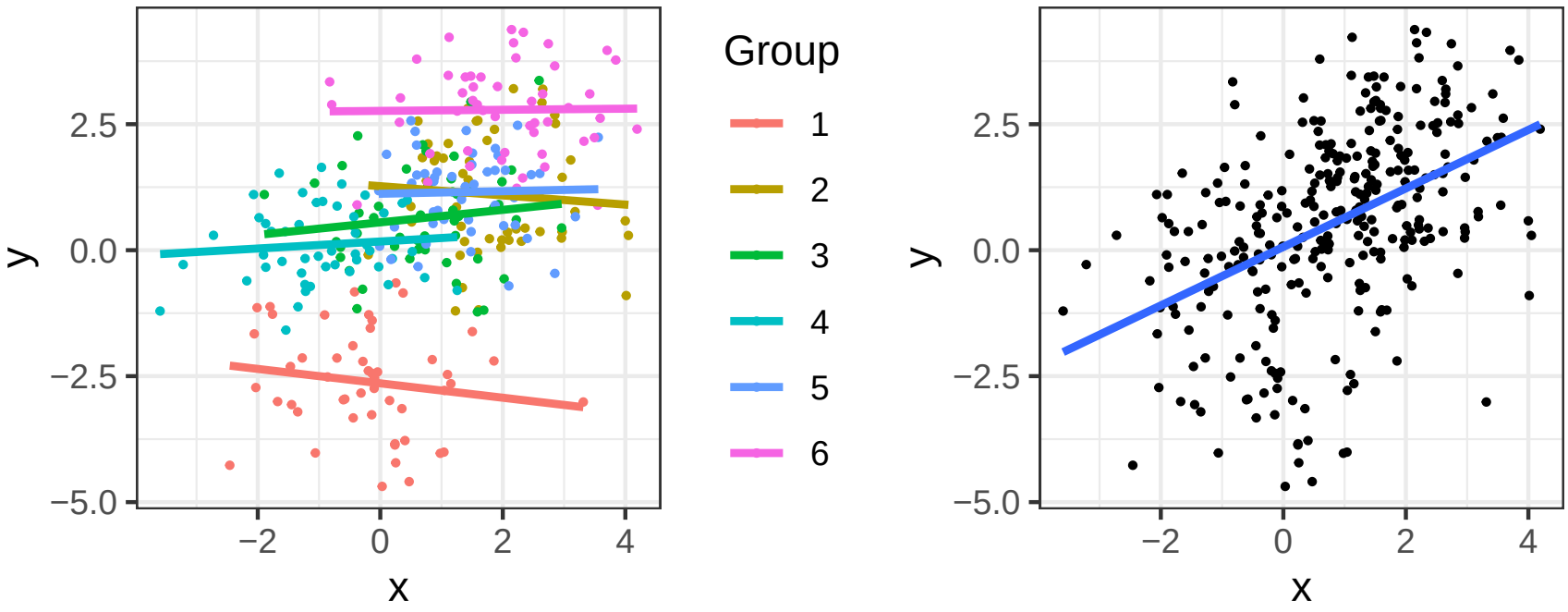
Question: What would an appropriate linear mixed model for this data and research question look like?

Warmup

```
Random effects:
 Groups       Name         Variance Std.Dev.
 neighborhood (Intercept) 1048     32.37
 Residual                 6762     82.23
Number of obs: 1561, groups:  neighborhood, 43

Fixed effects:
                     Estimate Std. Error t value
(Intercept)            27.275     30.547   0.893
overall_satisfaction   14.809      6.303   2.350
```

Last time: class activity



Linear regression: $\widehat{\beta}_1 = 0.5806,\ \widehat{SE}(\widehat{\beta}_1) = 0.066$

LMM: $\widehat{\beta}_1 = 0.00207,\ \widehat{SE}(\widehat{\beta}_1) = 0.0546$

Issues with ignoring group structure

- Estimating the wrong coefficients
- Standard errors are too small
- Inflated type I error rates for hypothesis tests

Models with a random intercept

Class activity

Work on the class activity (handout).

Models with a random intercept

Intra-class correlation

$$Y_{ij} = u_i + \mathbf{x}_{ij}^T \beta + \varepsilon_{ij}$$

The **intra-class correlation coefficient** (ICC) is the correlation between two observations within the same group i:

$$\rho_{\text{group}} := Cor(Y_{ij}, Y_{ik} | \mathbf{x}_{ij}, \mathbf{x}_{ik}) = \frac{\sigma_u^2}{\sigma_u^2 + \sigma_\varepsilon^2} \quad (j \neq k)$$

```
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Parameter estimation